

Research Meeting

2026-08-24

Ren Hirata

Research to date

Background

EHRs contain extensive information, while data needed for care is spread across multiple screens

Goal

Bring together the information required for each clinical task and support efficient review

Approach

Identify the required data and presentation from the task, then create a UI suited to its purpose

Change how clinicians reach the information they need

Fixed screens

Required information is spread across screens and records.

Users must remember where it is and how to reach it.

Task-aligned screens

The system gathers the data needed for the question at hand.

Lists, tables, and trends appear together for the same clinical context.

Recent progress and CHI 2027 submission

User evaluation

1 Discussing potential collaborators through Dr. Yabe



2 Starting after ethics and study requirements are met

CHI 2027 Papers

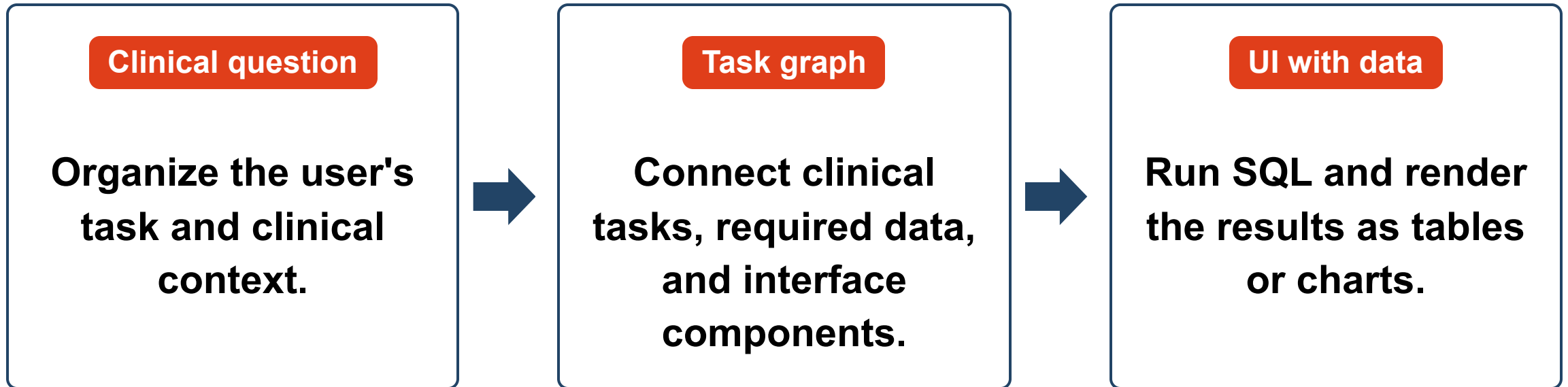
1 Build evaluation data from EHRSQL, which pairs clinical questions with reference SQL



2 Detect, locate, and repair gaps across questions, SQL, query results, and screens

The first submission will focus on technical evaluation; human-subject evaluation will follow as future validation

Connect tasks, data, and presentation to build the UI



Current implementation and future generation

Current

We manually create structural JSON from interviews.

The system reads the task graph and renders the UI with rules.

Future

A custom model will interpret the user's request.

It will construct the proposed task graph and update the UI.

Preoperative anesthesia scenario developed with Dr. Ito

We organized the clinical setting and review tasks based on interviews with Dr. Ito

Clinical setting

Before surgery, review the patient's background and prepare anesthesia precautions and explanations

Review tasks

- **Patient background and planned surgery**
- **Social history and allergies**
- **Medical history, treatment, and laboratory results**
- **Anesthesia history and anesthesia risks**

Preoperative anesthesia clinic UI

Interactive EHR

表示するシナリオ

麻酔科術前外来

UI生成・編集ツール

INTERACTIVE EHR

麻酔科術前外来

伊藤 太郎 / 72歳 / 男性 / ID ITO_PREOP_001

研究用・合成データ 診療判断には使用できません 情報源: 患者基本、患者プロフィール、患者アレルギー情報 ほか5件

最終データ日時: 2026-06-20 画面構成: 標準デモ 画面更新: 2026-08-24 07:19

患者情報の確認 手術の確認 検査結果の確認

情報源: 患者基本、患者プロフィール、患者アレルギー情報 ほか3件 | 最終データ日時: 2026-05-01

患者基本情報

生活歴・アレルギー

- 患者ID: ITO_PREOP_001
- 匿名ID: ITO_ANON_001
- 年齢: 72
- 性別: 男性
- 身長: 165 cm
- 体重: 64 kg
- BMI: 23.5

- 喫煙歴: 禁煙
- 喫煙開始年齢: 20
- 喫煙中止年齢: 55
- 飲酒歴: 機会飲酒
- アレルギー: ペニシリン
- アレルギー症状: 発疹
- コメント: 術前に抗菌薬選択要確認

Patient information

Interactive EHR

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患者情報の確認 手術の確認 検査結果の確認

情報源: 検体検査結果 | 最終データ日時: 2026-05-20

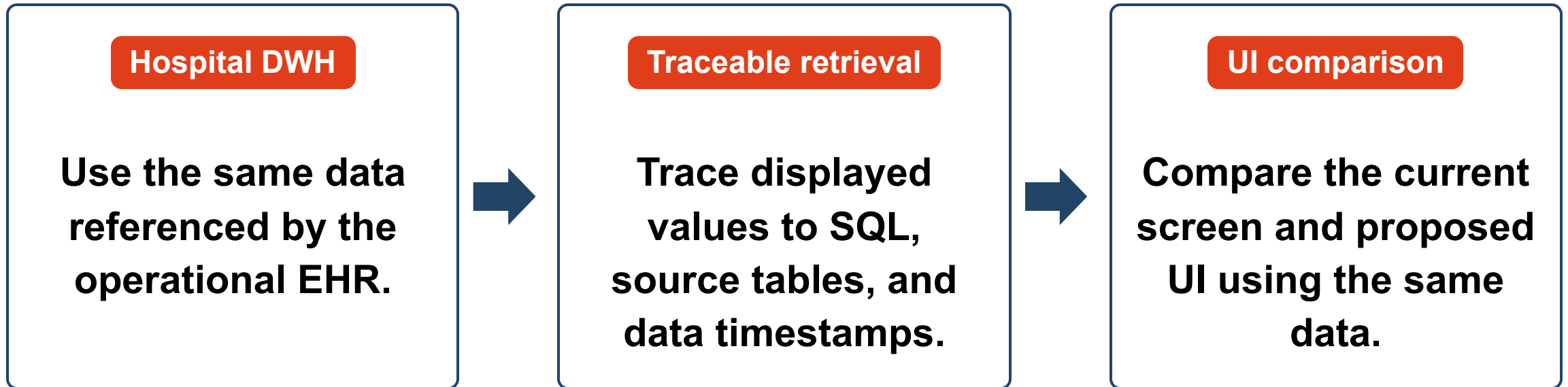
最新検査値一覧

区分	検査項目	最新値	検査日
血算	Hb	11.5 g/dL	2026-05-20
血算	WBC	7.1 ×10 ³ /μL	2026-05-20
血算	Plt	21.8 ×10 ⁴ /μL	2026-05-20
腎機能	Cr	1.3 mg/dL	2026-05-20
腎機能	eGFR	42.0 mL/min/1.73m ²	2026-05-20
電解質	K	4.8 mEq/L	2026-05-20

Laboratory results

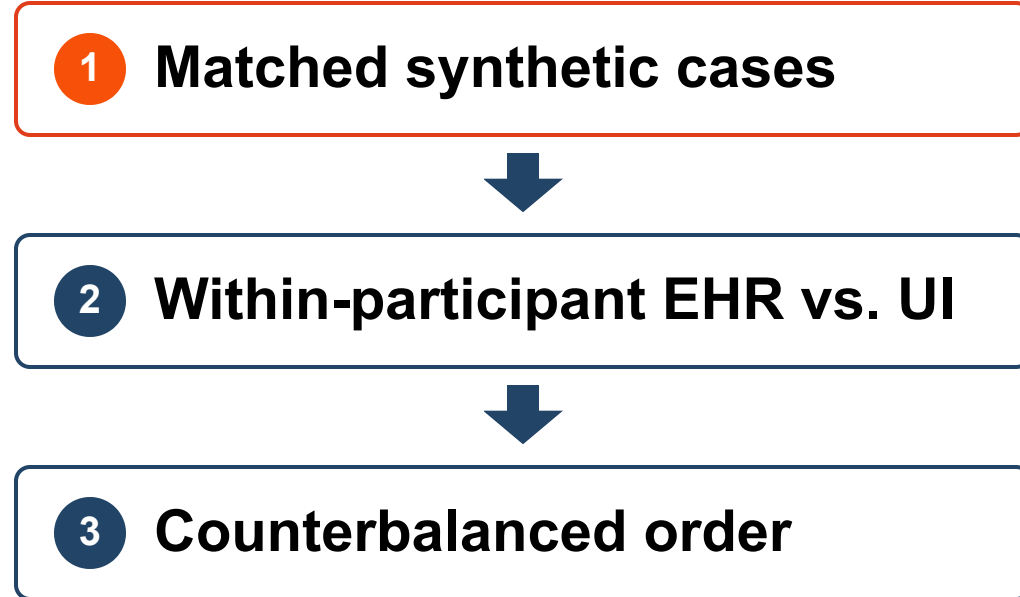
Synthetic data only. The patient and timestamp remain fixed while the review target changes

Compare screens using the same DWH data

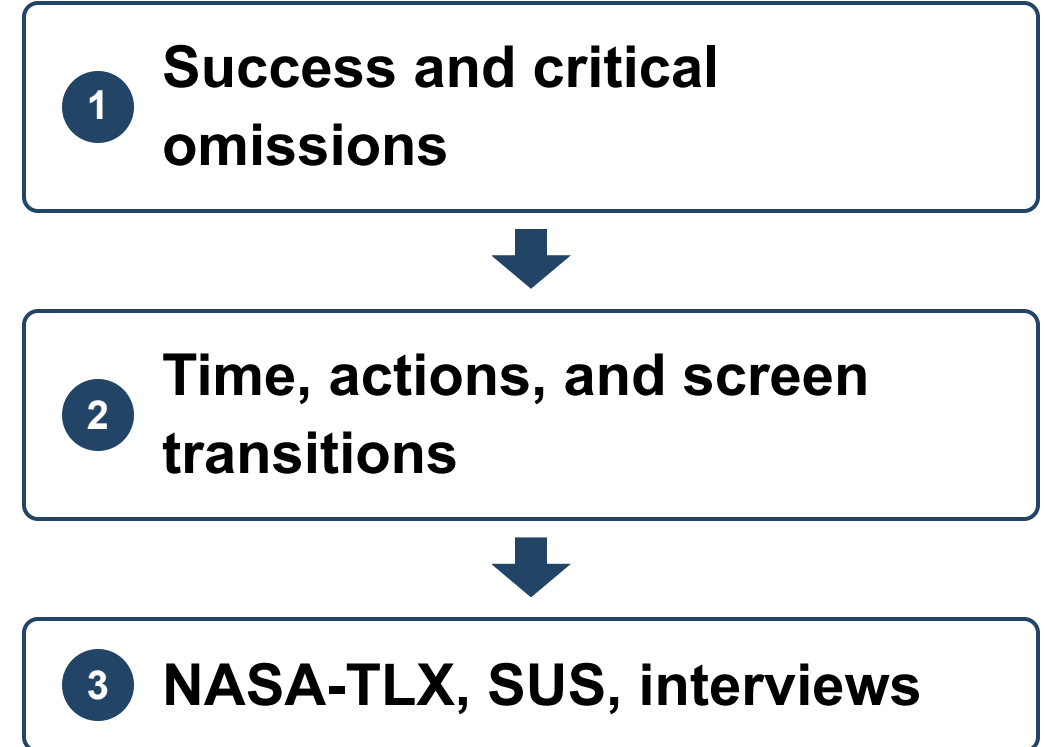


Planned user evaluation

Comparison design



Measures



Begin after expert review, ethics approval, and institutional authorization

Feedback requested from clinicians

Review it as a clinical interface

- **Does it include the information needed to prevent critical omissions?**
- **Are the information groups and review order natural?**
- **Can the source, timestamp, and retrieval conditions support verification?**
- **Are the comparison conditions with the current EHR fair?**